XRPL Validator — Daily Engineering Report · 2026-04-07

Period: 2026-04-07 00:00 to 24:00 US Eastern (2026-04-07 04:00 to 2026-04-08 04:00 UTC). Generated 2026-10-21 00:47 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: data.xrpl.org agreement scores. Not available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); signing records (they begin 2026-09-19); Step 1 receipt records (they begin 2026-08-29); Steps 2–5 receipt records (they begin 2026-10-09); deployment records (they begin 2026-09-11); xrpl-validator merge records (they begin 2026-07-04); amendment data.

Status

Not assessed — no signing records for this day.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** no signing records for this day (they begin 2026-09-19).
- **Native engine:** no receipt records for this day (they begin 2026-08-29).
- **Changes:** 0 deployments, 15 commits (0 with a recorded merge into main; xrpl-validator merge records begin 2026-07-04).

Service levels

No signing records for this day (they begin 2026-09-19).

External agreement, data.xrpl.org (UTC day 2026-04-07): 0.98327 (341 missed of 20,381). *data.xrpl.org scores UTC days from the validations its crawlers received.*

Native engine conformance

No receipt records for this day (Step 1 records begin 2026-08-29, Steps 2–5 records 2026-10-09).

Live counters are kept only for the current run; none exist for a past day.

Receipt register

No receipt records for this day (Step 1 records begin 2026-08-29).

Incidents

Not assessed: no signing records for this day.

Change log

Deployments

No deployment records before 2026-09-11.

Merged into main

Merge records begin 2026-07-04; this day's commits are listed by commit date below.

Findings fixed

No engine findings merged into main this day.

Other engineering changes

- **feat(ffi): Stage 2 — shadow state hash from FFI mutations** (`1a7c435`, xrpl-validator) After FFI applies every tx in a ledger, the accumulated mutations (overlay) are now returned from apply_ledger_in_order. The caller can compute a "shadow" state hash by cloning the current FlatHasher, applying the overlay (Created/Modified/Deleted SLEs), and computing the Merkle root — then comparing it to rippled's account_hash.
- **fix(ffi): shadow hash must snapshot hasher BEFORE process_ledger writes — fixes double-apply mismatch** (`a2feb29`, xrpl-validator)
- **fix(ffi): merge protocol-level state changes into shadow overlay — covers singletons + pseudo-tx effects** (`bf46baf`, xrpl-validator)
- **fix(ffi): shadow hash uses RPC canonical node_binary — fixes serialization mismatch** (`1eaf2a1`, xrpl-validator)
- **fix(ffi): move singleton insertion before shadow hash — shadow overlay now covers all state changes** (`b9d476a`, xrpl-validator)
- **feat(ui): shadow hash counter on hero banner** (`3f51bb2`, xrpl-validator)
- **feat(ui): show shadow hash hex comparison on hero — ours vs network with match indicator** (`9efb8cc`, xrpl-validator)
- **fix(ffi): terPRE_SEQ recovery — fetch sender AccountRoot at current ledger to break cascade** (`c88e6a8`, xrpl-validator)

- **fix(ffi): skip shadow hash until 5 consecutive state-hash matches — prevents false mismatch during settle** (`915782c`, xrpl-validator)
- **feat: serve static HTML from disk — no rebuild needed for frontend changes** (`021d687`, xrpl-validator)
- **feat: network topology page (PCB style) + catch-all static route for future pages** (`e4f2140`, xrpl-validator)
- **fix: axum v0.7 path syntax {name} not :name** (`9aaca2b`, xrpl-validator)

Docs and operations

- **debug: add overlay size logging to shadow hash mismatch** (`27261cf`, xrpl-validator)
- **docs: test coverage proof — unit tests + live production metrics** (`e9dcf1e`, xrpl-validator)
- **debug: add terPRE_SEQ recovery logging + fallback to ledger-1** (`6c18f6e`, xrpl-validator)

Independence program

No independence record for UTC day 2026-04-07; records begin 2026-10-20.

Network

No ledger records for this day: the reference node holds ledgers from 2026-10-04 22:52 UTC (#107,436,647).

Amendment watch

No amendment records for this day.

Open items

None.

Corrections to earlier reports

None.

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
04-07 10:10	—	xrpl-validator	`1a7c435`	feat(ffi): Stage 2 — shadow state hash from FFI mutations
04-07 10:16	—	xrpl-validator	`a2feb29`	fix(ffi): shadow hash must snapshot hasher BEFORE process_ledger writes — fixes double-apply mismatch
04-07 10:23	—	xrpl-validator	`bf46baf`	fix(ffi): merge protocol-level state changes into shadow overlay — covers singletons + pseudo-tx effects

Committed	Merged into main	Repo	Commit	Subject
04-07 10:29	—	xrpl-validator	`27261cf`	debug: add overlay size logging to shadow hash mismatch
04-07 10:37	—	xrpl-validator	`1eaf2a1`	fix(ffi): shadow hash uses RPC canonical node_binary — fixes serialization mismatch
04-07 10:43	—	xrpl-validator	`b9d476a`	fix(ffi): move singleton insertion before shadow hash — shadow overlay now covers all state changes
04-07 10:56	—	xrpl-validator	`e9dcf1e`	docs: test coverage proof — unit tests + live production metrics
04-07 11:52	—	xrpl-validator	`3f51bb2`	feat(ui): shadow hash counter on hero banner
04-07 12:00	—	xrpl-validator	`9efb8cc`	feat(ui): show shadow hash hex comparison on hero — ours vs network with match indicator
04-07 13:40	—	xrpl-validator	`c88e6a8`	fix(ffi): terPRE_SEQ recovery — fetch sender AccountRoot at current ledger to break cascade
04-07 14:06	—	xrpl-validator	`915782c`	fix(ffi): skip shadow hash until 5 consecutive state-hash matches — prevents false mismatch during settle
04-08 01:58	—	xrpl-validator	`6c18f6e`	debug: add terPRE_SEQ recovery logging + fallback to ledger-1
04-08 02:30	—	xrpl-validator	`021d687`	feat: serve static HTML from disk — no rebuild needed for frontend changes
04-08 02:47	—	xrpl-validator	`e4f2140`	feat: network topology page (PCB style) + catch-all static route for future pages
04-08 02:50	—	xrpl-validator	`9aaca2b`	fix: axum v0.7 path syntax {name} not :name

Times in UTC. A commit appears on the day it was committed to main or merged into it.